

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 1000.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 396

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 33

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 152222867664.8708

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 249999.99972411105

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 28320476.091023352

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 88134.44313115414

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.499999999930564

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 1.0521508211085393

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition -73894.45495589092

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 0.0811

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 0.8846

Parameter bounds [0.000E+000; 3.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 0.0000

Parameter bounds [0.000E+000; 3.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.0000

Parameter bounds [0.000E+000; 1.000E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999999936317.121

Parameter bounds [1.000E+000; 1.000E+013]

Activation energy for condensed phase reactions 183793.83206061024

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 128329227111.24438

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 214486.11137722578

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 1.1218870136950905

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 7.123251921524469E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 2.2429027909374662E-12

Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 268185.7536857157

Parameter bounds [1.000E+000; 1.000E+013]

Activation energy for condensed phase reactions 50000.000011454846

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 16407654.742608069

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 50000.00027831996

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 1.988813938735339

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 1.1900718349466088E-06

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 1.649928813633361E-09

Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999966726912.803

Parameter bounds [1.000E+000; 1.000E+013]

Activation energy for condensed phase reactions 183793.80723672872

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 7866507.64260538

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 50000.00000809804

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 1.614292234882693

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 9.218575972615838E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 9.554992103530617E-06

Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.0583

Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s

Calculated burning rate 7.78 mm/s

First Conditional Fuel

Surface temperature 821.68 K

Linear burning rate 12.49 mm/s

Mass decomposition rate 20.72 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 1.47e+07 W/m²

Kinetic flame height 19.57 μm
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature 2,264.34 K
Binder sublimation heat flux density 1.47e+07 W/m^2
Surface heat balance error -0.0024 W/m^2

Secondary Conditional Fuel

Surface temperature 785.08 K
Linear burning rate 3.56 mm/s
Mass decomposition rate 5.91 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.39e+06 W/m^2
Kinetic flame height 66.09 μm
Average kinetic flame density 2.52 kg/m^3
Average kinetic flame temperature 1,575.04 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 4.76e+06 W/m^2
Kinetic flame height 16.93 μm
Average kinetic flame density 3.34 kg/m^3
Average kinetic flame temperature 1,188.04 K
Heat flux density due to metal combustion, 1.6e+06 W/m^2
Effective thermal conductivity 0.42 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 0.015 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.35e-04 m
Pore diameter 9.56e-06 m
Diffusion flame heat flux density 67,363.39 W/m^2
Diffusion flame height 4,184.64 μm
Heat flux density to surface within skeletal structure 2.28e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.53e+06 W/m^2
Total heat flux density to surface 3.88e+06 W/m^2
Binder sublimation heat flux density 3.88e+06 W/m^2
Surface heat balance error -7.13e-04 W/m^2

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s
Calculated burning rate 20.86 mm/s

First Conditional Fuel

Surface temperature 878.86 K
Linear burning rate 71.89 mm/s
Mass decomposition rate 119.27 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 9.51e+07 W/m^2
Kinetic flame height 2.97 μm
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature 2,292.93 K
Binder sublimation heat flux density 9.51e+07 W/m^2
Surface heat balance error -0.011 W/m^2

Secondary Conditional Fuel

Surface temperature 811.7 K
Linear burning rate 8.97 mm/s
Mass decomposition rate 14.88 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 4.27e+06 W/m^2
Kinetic flame height 36.37 μm
Average kinetic flame density 10.13 kg/m^3
Average kinetic flame temperature 1,588.35 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.82e+07 W/m^2
Kinetic flame height 4.28 μm

Average kinetic flame density 13.4 kg/m^3
Average kinetic flame temperature $1,201.35 \text{ K}$
Heat flux density due to metal combustion, $3.43\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.015 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$
Skeleton layer thickness $5.96\text{e-}05 \text{ m}$
Pore diameter $9.56\text{e-}06 \text{ m}$
Diffusion flame heat flux density $361,757.22 \text{ W/m}^2$
Diffusion flame height $771.87 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $7.15\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $2.86\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $1.04\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.04\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0015 W/m^2

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s
Calculated burning rate 27.63 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 129.79 mm/s
Mass decomposition rate $215.32 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.79\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.57 \text{ }\mu\text{m}$
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature $2,303.5 \text{ K}$
Binder sublimation heat flux density $1.79\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.041 W/m^2

Secondary Conditional Fuel

Surface temperature 819.74 K
Linear burning rate 11.72 mm/s
Mass decomposition rate $19.44 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $5.42\text{e}+06 \text{ W/m}^2$
Kinetic flame height $28.51 \text{ }\mu\text{m}$
Average kinetic flame density 16.2 kg/m^3
Average kinetic flame temperature $1,592.37 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.99\text{e}+07 \text{ W/m}^2$
Kinetic flame height $2.58 \text{ }\mu\text{m}$
Average kinetic flame density 21.4 kg/m^3
Average kinetic flame temperature $1,205.37 \text{ K}$
Heat flux density due to metal combustion, $4.27\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.015 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$
Skeleton layer thickness $4.71\text{e-}05 \text{ m}$
Pore diameter $9.56\text{e-}06 \text{ m}$
Diffusion flame heat flux density $552,460.12 \text{ W/m}^2$
Diffusion flame height $503.97 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $9.28\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $3.95\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $1.38\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.38\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0016 W/m^2

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
Calculated burning rate 16.67 mm/s

First Conditional Fuel

Surface temperature 821.68 K
Linear burning rate 12.49 mm/s
Mass decomposition rate $20.72 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.47\text{e}+07 \text{ W/m}^2$
Kinetic flame height 19.57 μm
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature 2,264.34 K
Binder sublimation heat flux density $1.47\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0024 W/m^2

Secondary Conditional Fuel

Surface temperature 821.68 K
Linear burning rate 12.49 mm/s
Mass decomposition rate $20.72 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $711,118.64 \text{ W/m}^2$
Kinetic flame height 217.03 μm
Average kinetic flame density 2.49 kg/m^3
Average kinetic flame temperature 1,593.34 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $8.01\text{e}+06 \text{ W/m}^2$
Kinetic flame height 22.47 μm
Average kinetic flame density 2.3 kg/m^3
Average kinetic flame temperature 1,722.34 K
Heat flux density due to metal combustion, $3.4\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.32 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.095 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E-}08$
Skeleton layer thickness $4.45\text{e-}05 \text{ m}$
Pore diameter $9.56\text{e-}06 \text{ m}$
Diffusion flame heat flux density $1.07\text{e}+07 \text{ W/m}^2$
Diffusion flame height 26.7 μm
Heat flux density to surface within skeletal structure $3.57\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $488,602.16 \text{ W/m}^2$
Total heat flux density to surface $1.47\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.47\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0016 W/m^2

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 36 mm/s

First Conditional Fuel

Surface temperature 878.86 K
Linear burning rate 71.89 mm/s
Mass decomposition rate $119.27 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $9.51\text{e}+07 \text{ W/m}^2$
Kinetic flame height 2.97 μm
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature 2,292.93 K
Binder sublimation heat flux density $9.51\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.011 W/m^2

Secondary Conditional Fuel

Surface temperature 833.57 K

Linear burning rate 18.33 mm/s
Mass decomposition rate $30.41 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.14\text{e}+06 \text{ W/m}^2$
Kinetic flame height $71.5 \mu\text{m}$
Average kinetic flame density 10.06 kg/m^3
Average kinetic flame temperature 1,599.28 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.23\text{e}+07 \text{ W/m}^2$
Kinetic flame height $3.42 \mu\text{m}$
Average kinetic flame density 9.31 kg/m^3
Average kinetic flame temperature 1,728.28 K
Heat flux density due to metal combustion, $4.65\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.32 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.095 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness $3.17\text{e}-05 \text{ m}$
Pore diameter $9.56\text{e}-06 \text{ m}$

Diffusion flame heat flux density $7.25\text{e}+06 \text{ W/m}^2$
Diffusion flame height $39.18 \mu\text{m}$
Heat flux density to surface within skeletal structure $1.33\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.64\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $2.22\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $2.22\text{e}+07 \text{ W/m}^2$
Surface heat balance error $-8.08\text{e}-04 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s
Calculated burning rate 42.98 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 129.79 mm/s
Mass decomposition rate $215.32 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.79\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.57 \mu\text{m}$
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density $1.79\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.041 W/m^2

Secondary Conditional Fuel

Surface temperature 837.17 K
Linear burning rate 20.54 mm/s
Mass decomposition rate $34.08 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $3.15\text{e}+06 \text{ W/m}^2$
Kinetic flame height $48.48 \mu\text{m}$
Average kinetic flame density 16.11 kg/m^3
Average kinetic flame temperature 1,601.08 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.79 \mu\text{m}$
Average kinetic flame density 14.91 kg/m^3
Average kinetic flame temperature 1,730.08 K
Heat flux density due to metal combustion, $5.11\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.32 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.095 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness $2.87\text{e-}05$ m
Pore diameter $9.56\text{e-}06$ m
Diffusion flame heat flux density $6.46\text{e+}06$ W/m²
Diffusion flame height 43.9 μm
Heat flux density to surface within skeletal structure $1.59\text{e+}07$ W/m²
Heat flux density to surface outside skeletal structure $2.67\text{e+}06$ W/m²
Total heat flux density to surface $2.5\text{e+}07$ W/m²
Binder sublimation heat flux density $2.5\text{e+}07$ W/m²
Surface heat balance error -0.0069 W/m²

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s
Calculated burning rate 9.74 mm/s

First Conditional Fuel

Surface temperature 821.68 K
Linear burning rate 12.49 mm/s
Mass decomposition rate 20.72 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface $1.47\text{e+}07$ W/m²
Kinetic flame height 19.57 μm
Average kinetic flame density 1.75 kg/m³
Average kinetic flame temperature $2,264.34$ K
Binder sublimation heat flux density $1.47\text{e+}07$ W/m²
Surface heat balance error -0.0024 W/m²

Secondary Conditional Fuel

Surface temperature 782.5 K
Linear burning rate 3.25 mm/s
Mass decomposition rate 5.39 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $5,759.35$ W/m²
Kinetic flame height $4,505.74$ μm
Average kinetic flame density 4.35 kg/m³
Average kinetic flame temperature 912.25 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.59\text{e+}06$ W/m²
Kinetic flame height 28.75 μm
Average kinetic flame density 3.92 kg/m³
Average kinetic flame temperature $1,011.75$ K
Heat flux density due to metal combustion, $5\text{e+}06$ W/m²
Effective thermal conductivity 1.42 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.0058 W/(m·K)
Conductive thermal conductivity balance error $9.240271880983641\text{E-}10$
Skeleton layer thickness $1.47\text{e-}04$ m
Pore diameter $9.56\text{e-}06$ m
Diffusion flame heat flux density $26,949.63$ W/m²
Diffusion flame height $10,380.48$ μm
Heat flux density to surface within skeletal structure $3.49\text{e+}06$ W/m²
Heat flux density to surface outside skeletal structure $2,713.99$ W/m²
Total heat flux density to surface $3.52\text{e+}06$ W/m²
Binder sublimation heat flux density $3.52\text{e+}06$ W/m²
Surface heat balance error $1.19\text{e-}04$ W/m²

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 25.93 mm/s

First Conditional Fuel

Surface temperature 878.86 K

Linear burning rate 71.89 mm/s
Mass decomposition rate $119.27 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $9.51\text{e}+07 \text{ W/m}^2$
Kinetic flame height $2.97 \mu\text{m}$
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature 2,292.93 K
Binder sublimation heat flux density $9.51\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.011 W/m^2

Secondary Conditional Fuel

Surface temperature 807.5 K
Linear burning rate 7.79 mm/s
Mass decomposition rate $12.92 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $10,922.07 \text{ W/m}^2$
Kinetic flame height $2,147.05 \mu\text{m}$
Average kinetic flame density 17.41 kg/m^3
Average kinetic flame temperature 924.75 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.35\text{e}+06 \text{ W/m}^2$
Kinetic flame height $6.82 \mu\text{m}$
Average kinetic flame density 15.72 kg/m^3
Average kinetic flame temperature 1,024.25 K
Heat flux density due to metal combustion, $1.03\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $1.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.0058 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $6.76\text{e}-05 \text{ m}$
Pore diameter $9.56\text{e}-06 \text{ m}$

Diffusion flame heat flux density $161,573.75 \text{ W/m}^2$
Diffusion flame height $1,715.94 \mu\text{m}$
Heat flux density to surface within skeletal structure $8.75\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $5,188.21 \text{ W/m}^2$
Total heat flux density to surface $8.92\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $8.92\text{e}+06 \text{ W/m}^2$
Surface heat balance error $6.37\text{e}-04 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 36.04 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 129.79 mm/s
Mass decomposition rate $215.32 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.79\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.57 \mu\text{m}$
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density $1.79\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.041 W/m^2

Secondary Conditional Fuel

Surface temperature 816.74 K
Linear burning rate 10.61 mm/s
Mass decomposition rate $17.61 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $13,313.43 \text{ W/m}^2$
Kinetic flame height $1,691.95 \mu\text{m}$
Average kinetic flame density 27.76 kg/m^3
Average kinetic flame temperature 929.37 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $9.96 \times 10^6 \text{ W/m}^2$
Kinetic flame height $4.26 \text{ }\mu\text{m}$
Average kinetic flame density 25.07 kg/m^3
Average kinetic flame temperature $1,028.87 \text{ K}$
Heat flux density due to metal combustion, $1.33 \times 10^7 \text{ W/m}^2$
Effective thermal conductivity $1.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.0058 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641 \times 10^{-10}$
Skeleton layer thickness $5.14 \times 10^{-5} \text{ m}$
Pore diameter $9.56 \times 10^{-6} \text{ m}$
Diffusion flame heat flux density $253,460.8 \text{ W/m}^2$
Diffusion flame height $1,090.21 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $1.21 \times 10^7 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $6,366.98 \text{ W/m}^2$
Total heat flux density to surface $1.24 \times 10^7 \text{ W/m}^2$
Binder sublimation heat flux density $1.24 \times 10^7 \text{ W/m}^2$
Surface heat balance error -0.0014 W/m^2

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s
Calculated burning rate 14.23 mm/s

First Conditional Fuel

Surface temperature 638.27 K
Linear burning rate 13.09 mm/s
Mass decomposition rate $21.71 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $9.48 \times 10^6 \text{ W/m}^2$
Kinetic flame height $32.38 \text{ }\mu\text{m}$
Average kinetic flame density 1.83 kg/m^3
Average kinetic flame temperature $2,172.64 \text{ K}$
Binder sublimation heat flux density $9.48 \times 10^6 \text{ W/m}^2$
Surface heat balance error $-7.52 \times 10^{-4} \text{ W/m}^2$

Secondary Conditional Fuel

Surface temperature 600 K
Linear burning rate 7.17 mm/s
Mass decomposition rate $11.9 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $928,791.2 \text{ W/m}^2$
Kinetic flame height $190.03 \text{ }\mu\text{m}$
Average kinetic flame density 2.68 kg/m^3
Average kinetic flame temperature $1,482.5 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $7.3 \times 10^6 \text{ W/m}^2$
Kinetic flame height $13.57 \text{ }\mu\text{m}$
Average kinetic flame density 3.62 kg/m^3
Average kinetic flame temperature $1,095.5 \text{ K}$
Heat flux density due to metal combustion, $3.01 \times 10^6 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $2.66 \times 10^{-6} \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
Skeleton layer thickness $9.38 \times 10^{-5} \text{ m}$
Pore diameter $1.65 \times 10^{-9} \text{ m}$
Diffusion flame heat flux density $36,390.77 \text{ W/m}^2$
Diffusion flame height $8,425.21 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $3.9 \times 10^6 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure 577,657.69 W/m²
Total heat flux density to surface 4.51e+06 W/m²
Binder sublimation heat flux density 4.51e+06 W/m²
Surface heat balance error -2.71e-04 W/m²

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s
Calculated burning rate 37.91 mm/s

First Conditional Fuel

Surface temperature 779.46 K
Linear burning rate 72.1 mm/s
Mass decomposition rate 119.62 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 7.75e+07 W/m²
Kinetic flame height 3.78 μm
Average kinetic flame density 7.18 kg/m³
Average kinetic flame temperature 2,243.23 K
Binder sublimation heat flux density 7.75e+07 W/m²
Surface heat balance error 0.003 W/m²

Secondary Conditional Fuel

Surface temperature 656.72 K
Linear burning rate 17.05 mm/s
Mass decomposition rate 28.29 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.85e+06 W/m²
Kinetic flame height 92.34 μm
Average kinetic flame density 10.65 kg/m³
Average kinetic flame temperature 1,510.86 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 5.12e+07 W/m²
Kinetic flame height 1.83 μm
Average kinetic flame density 14.32 kg/m³
Average kinetic flame temperature 1,123.86 K
Heat flux density due to metal combustion, 5.95e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.66e-06 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.36e-05 m
Pore diameter 1.65e-09 m
Diffusion flame heat flux density 205,147.73 W/m²
Diffusion flame height 1,466.88 μm
Heat flux density to surface within skeletal structure 1.14e+07 W/m²
Heat flux density to surface outside skeletal structure 1.48e+06 W/m²
Total heat flux density to surface 1.31e+07 W/m²
Binder sublimation heat flux density 1.31e+07 W/m²
Surface heat balance error 5.6e-04 W/m²

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s
Calculated burning rate 52.9 mm/s

First Conditional Fuel

Surface temperature 843.54 K
Linear burning rate 129.57 mm/s
Mass decomposition rate 214.95 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.6e+08 W/m²
Kinetic flame height 1.79 μm
Average kinetic flame density 11.34 kg/m³
Average kinetic flame temperature 2,275.27 K
Binder sublimation heat flux density 1.6e+08 W/m²

Surface heat balance error -0.0057 W/m²

Secondary Conditional Fuel

Surface temperature 679.79 K

Linear burning rate 23.26 mm/s

Mass decomposition rate 38.59 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.3e+06 W/m²

Kinetic flame height 73.31 μm

Average kinetic flame density 16.95 kg/m³

Average kinetic flame temperature 1,522.39 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 9.67e+07 W/m²

Kinetic flame height 0.94 μm

Average kinetic flame density 22.72 kg/m³

Average kinetic flame temperature 1,135.39 K

Heat flux density due to metal combustion, 7.55e+06 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.66e-06 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 3.31e-05 m

Pore diameter 1.65e-09 m

Diffusion flame heat flux density 298,472.27 W/m²

Diffusion flame height 1,000.5 μm

Heat flux density to surface within skeletal structure 1.7e+07 W/m²

Heat flux density to surface outside skeletal structure 1.92e+06 W/m²

Total heat flux density to surface 1.92e+07 W/m²

Binder sublimation heat flux density 1.92e+07 W/m²

Surface heat balance error 0.0012 W/m²

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s

Calculated burning rate 5.97 mm/s

First Conditional Fuel

Surface temperature 821.68 K

Linear burning rate 12.49 mm/s

Mass decomposition rate 20.72 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 1.47e+07 W/m²

Kinetic flame height 19.57 μm

Average kinetic flame density 1.75 kg/m³

Average kinetic flame temperature 2,264.34 K

Binder sublimation heat flux density 1.47e+07 W/m²

Surface heat balance error 2.36e-04 W/m²

Secondary Conditional Fuel

Surface temperature 777.04 K

Linear burning rate 2.66 mm/s

Mass decomposition rate 4.42 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3.17e+06 W/m²

Kinetic flame height 50.1 μm

Average kinetic flame density 2.53 kg/m³

Average kinetic flame temperature 1,571.02 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3,169.53 W/m²

Kinetic flame height 25,680.83 μm

Average kinetic flame density 3.35 kg/m³

Average kinetic flame temperature 1,184.02 K

Heat flux density due to metal combustion, $1.56 \times 10^6 \text{ W/m}^2$
 Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Radiative thermal conductivity $3.61 \times 10^{-9} \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
 Skeleton layer thickness $1.35 \times 10^{-4} \text{ m}$
 Pore diameter $2.24 \times 10^{-12} \text{ m}$
 Diffusion flame heat flux density $92,400.64 \text{ W/m}^2$
 Diffusion flame height $3,126.56 \text{ }\mu\text{m}$
 Heat flux density to surface within skeletal structure $405,112.08 \text{ W/m}^2$
 Heat flux density to surface outside skeletal structure $2.35 \times 10^6 \text{ W/m}^2$
 Total heat flux density to surface $2.85 \times 10^6 \text{ W/m}^2$
 Binder sublimation heat flux density $2.85 \times 10^6 \text{ W/m}^2$
 Surface heat balance error $3.06 \times 10^{-4} \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s
 Calculated burning rate 13.63 mm/s

First Conditional Fuel

Surface temperature 878.86 K
 Linear burning rate 71.89 mm/s
 Mass decomposition rate $119.27 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
 Kinetic flame heat flux density to surface $9.51 \times 10^7 \text{ W/m}^2$
 Kinetic flame height $2.97 \text{ }\mu\text{m}$
 Average kinetic flame density 7.02 kg/m^3
 Average kinetic flame temperature $2,292.93 \text{ K}$
 Binder sublimation heat flux density $9.51 \times 10^7 \text{ W/m}^2$
 Surface heat balance error 0.0079 W/m^2

Secondary Conditional Fuel

Surface temperature 798.68 K
 Linear burning rate 5.76 mm/s
 Mass decomposition rate $9.55 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $6.56 \times 10^6 \text{ W/m}^2$
 Kinetic flame height $23.88 \text{ }\mu\text{m}$
 Average kinetic flame density 10.18 kg/m^3
 Average kinetic flame temperature $1,581.84 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $8,277.42 \text{ W/m}^2$
 Kinetic flame height $9,572.02 \text{ }\mu\text{m}$
 Average kinetic flame density 13.47 kg/m^3
 Average kinetic flame temperature $1,194.84 \text{ K}$
 Heat flux density due to metal combustion, $2.96 \times 10^6 \text{ W/m}^2$
 Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Radiative thermal conductivity $3.61 \times 10^{-9} \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
 Skeleton layer thickness $6.83 \times 10^{-5} \text{ m}$
 Pore diameter $2.24 \times 10^{-12} \text{ m}$
 Diffusion flame heat flux density $579,033.19 \text{ W/m}^2$
 Diffusion flame height $495.19 \text{ }\mu\text{m}$
 Heat flux density to surface within skeletal structure $557,534.24 \text{ W/m}^2$
 Heat flux density to surface outside skeletal structure $5.33 \times 10^6 \text{ W/m}^2$
 Total heat flux density to surface $6.47 \times 10^6 \text{ W/m}^2$
 Binder sublimation heat flux density $6.47 \times 10^6 \text{ W/m}^2$
 Surface heat balance error $-9.77 \times 10^{-4} \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 18 mm/s

Calculated burning rate 18 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 129.79 mm/s
Mass decomposition rate $215.32 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $1.79\text{e}+08 \text{ W/m}^2$
Kinetic flame height 1.57 μm
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density $1.79\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.038 W/m^2

Secondary Conditional Fuel

Surface temperature 806.53 K
Linear burning rate 7.53 mm/s
Mass decomposition rate $12.5 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $8.31\text{e}+06 \text{ W/m}^2$
Kinetic flame height 18.76 μm
Average kinetic flame density 16.27 kg/m^3
Average kinetic flame temperature 1,585.76 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $11,365.58 \text{ W/m}^2$
Kinetic flame height 6,902.18 μm
Average kinetic flame density 21.52 kg/m^3
Average kinetic flame temperature 1,198.76 K
Heat flux density due to metal combustion, $3.7\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Radiative thermal conductivity $3.61\text{e}-09 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $5.38\text{e}-05 \text{ m}$
Pore diameter $2.24\text{e}-12 \text{ m}$

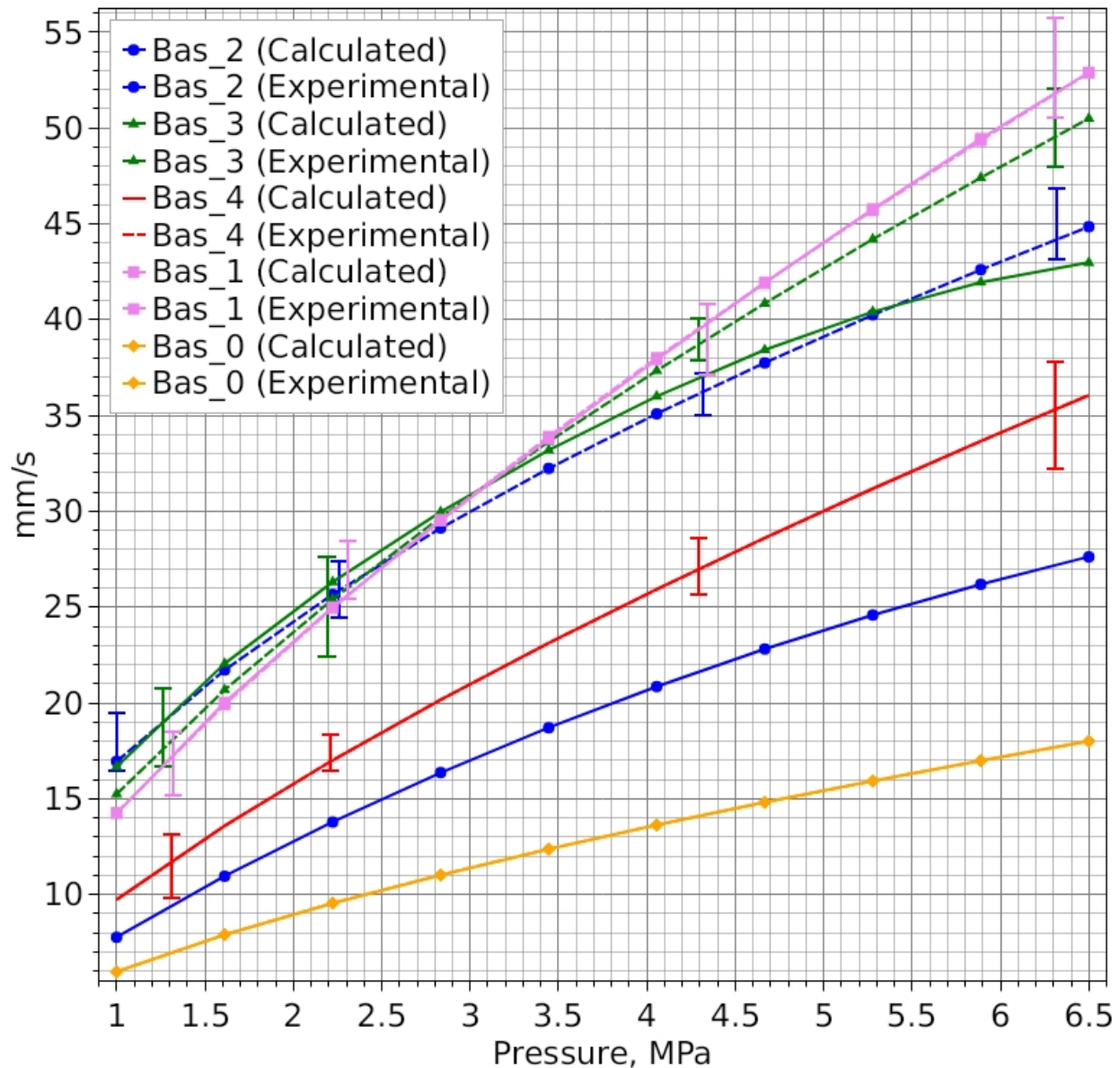
Diffusion flame heat flux density $882,549.33 \text{ W/m}^2$
Diffusion flame height 324 μm
Heat flux density to surface within skeletal structure $468,236.62 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $7.26\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $8.61\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $8.61\text{e}+06 \text{ W/m}^2$
Surface heat balance error -0.0021 W/m^2

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	7.78 mm/s		16.67 mm/s		9.74 mm/s		14.23 mm/s		5.97 mm/s	
Difference	9.17 mm/s		-1.43 mm/s		-0.016 mm/s		0.038 mm/s		-0.0084 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	12.49 mm/s	3.56 mm/s	12.49 mm/s	12.49 mm/s	12.49 mm/s	3.25 mm/s	13.09 mm/s	7.17 mm/s	12.49 mm/s	2.66 mm/s
Surface Temperature	821.68 K	785.08 K	821.68 K	821.68 K	821.68 K	782.5 K	638.27 K	600 K	821.68 K	777.04 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.47e+07 W/m ²		1.47e+07 W/m ²		1.47e+07 W/m ²		9.48e+06 W/m ²		1.47e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	4.76e+06 W/m ²		8.01e+06 W/m ²		1.59e+06 W/m ²		7.3e+06 W/m ²		3,169.53 W/m ²	
Kinetic Flame Height (Skeleton)	16.93 μm		22.47 μm		28.75 μm		13.57 μm		25,680.83 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.39e+06 W/m ²		711,118.64 W/m ²		5,759.35 W/m ²		928,791.2 W/m ²		3.17e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	66.09 μm		217.03 μm		4,505.74 μm		190.03 μm		50.1 μm	
Diffusion Flame Heat Flux Density	67,363.39 W/m ²		1.07e+07 W/m ²		26,949.63 W/m ²		36,390.77 W/m ²		92,400.64 W/m ²	
Diffusion Flame Height	4,184.64 μm		26.7 μm		10,380.48 μm		8,425.21 μm		3,126.56 μm	
Metal Combustion Heat Flux Density	1.6e+06 W/m ²		3.4e+06 W/m ²		5e+06 W/m ²		3.01e+06 W/m ²		1.56e+06 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	20.86 mm/s		36 mm/s		25.93 mm/s		37.91 mm/s		13.63 mm/s	
Difference	14.24 mm/s		1.34 mm/s		-0.011 mm/s		0.097 mm/s		-2.79e-04 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	71.89 mm/s	8.97 mm/s	71.89 mm/s	18.33 mm/s	71.89 mm/s	7.79 mm/s	72.1 mm/s	17.05 mm/s	71.89 mm/s	5.76 mm/s
Surface Temperature	878.86 K	811.7 K	878.86 K	833.57 K	878.86 K	807.5 K	779.46 K	656.72 K	878.86 K	798.68 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	9.51e+07 W/m ²		9.51e+07 W/m ²		9.51e+07 W/m ²		7.75e+07 W/m ²		9.51e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.82e+07 W/m ²		5.23e+07 W/m ²		6.35e+06 W/m ²		5.12e+07 W/m ²		8,277.42 W/m ²	
Kinetic Flame Height (Skeleton)	4.28 μm		3.42 μm		6.82 μm		1.83 μm		9,572.02 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	4.27e+06 W/m ²		2.14e+06 W/m ²		10,922.07 W/m ²		1.85e+06 W/m ²		6.56e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	36.37 μm		71.5 μm		2,147.05 μm		92.34 μm		23.88 μm	
Diffusion Flame Heat Flux Density	361,757.22 W/m ²		7.25e+06 W/m ²		161,573.75 W/m ²		205,147.73 W/m ²		579,033.19 W/m ²	
Diffusion Flame Height	771.87 μm		39.18 μm		1,715.94 μm		1,466.88 μm		495.19 μm	
Metal Combustion Heat Flux Density	3.43e+06 W/m ²		4.65e+06 W/m ²		1.03e+07 W/m ²		5.95e+06 W/m ²		2.96e+06 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	27.63 mm/s		42.98 mm/s		36.04 mm/s		52.9 mm/s		18 mm/s	
Difference	17.22 mm/s		7.52 mm/s		0.018 mm/s		-0.018 mm/s		-5.52e-04 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	129.79 mm/s	11.72 mm/s	129.79 mm/s	20.54 mm/s	129.79 mm/s	10.61 mm/s	129.57 mm/s	23.26 mm/s	129.79 mm/s	7.53 mm/s
Surface Temperature	900 K	819.74 K	900 K	837.17 K	900 K	816.74 K	843.54 K	679.79 K	900 K	806.53 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.79e+08 W/m ²		1.79e+08 W/m ²		1.79e+08 W/m ²		1.6e+08 W/m ²		1.79e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	2.99e+07 W/m ²		1e+08 W/m ²		9.96e+06 W/m ²		9.67e+07 W/m ²		11,365.58 W/m ²	
Kinetic Flame Height (Skeleton)	2.58 μm		1.79 μm		4.26 μm		0.94 μm		6,902.18 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	5.42e+06 W/m ²		3.15e+06 W/m ²		13,313.43 W/m ²		2.3e+06 W/m ²		8.31e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	28.51 μm		48.48 μm		1,691.95 μm		73.31 μm		18.76 μm	
Diffusion Flame Heat Flux Density	552,460.12 W/m ²		6.46e+06 W/m ²		253,460.8 W/m ²		298,472.27 W/m ²		882,549.33 W/m ²	
Diffusion Flame Height	503.97 μm		43.9 μm		1,090.21 μm		1,000.5 μm		324 μm	
Metal Combustion Heat Flux Density	4.27e+06 W/m ²		5.11e+06 W/m ²		1.33e+07 W/m ²		7.55e+06 W/m ²		3.7e+06 W/m ²	

Burning Rates - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341\text{E-}05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

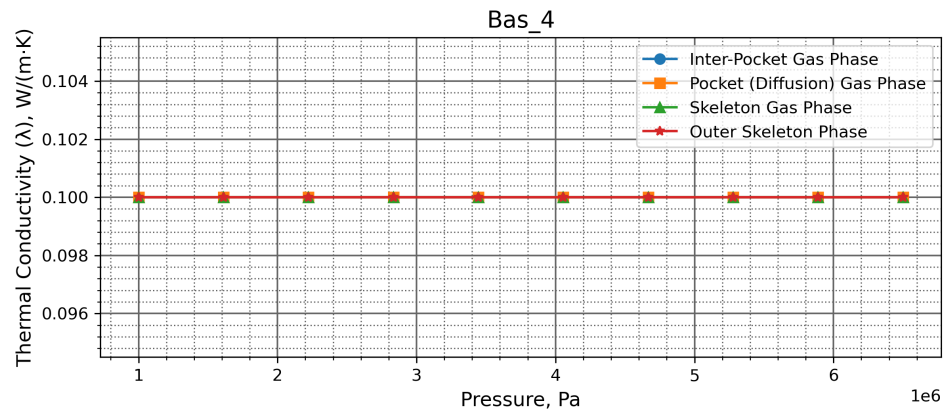
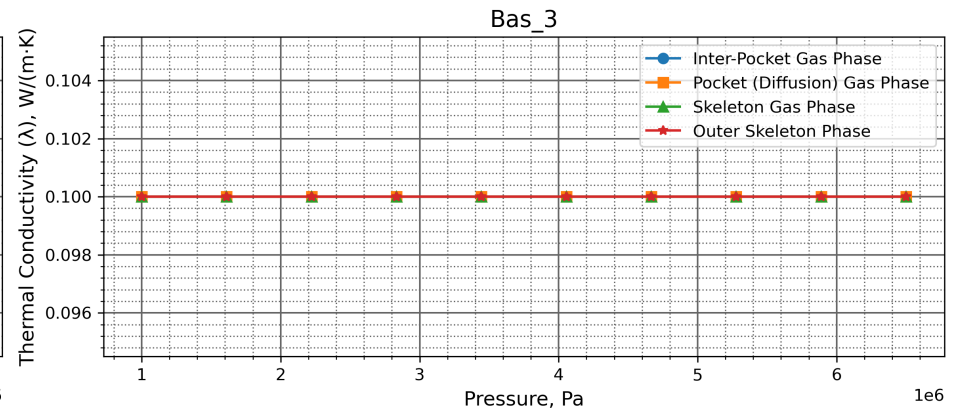
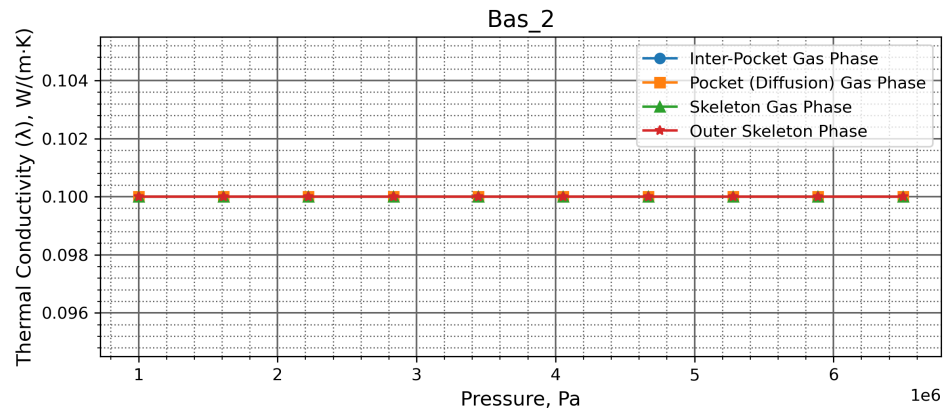
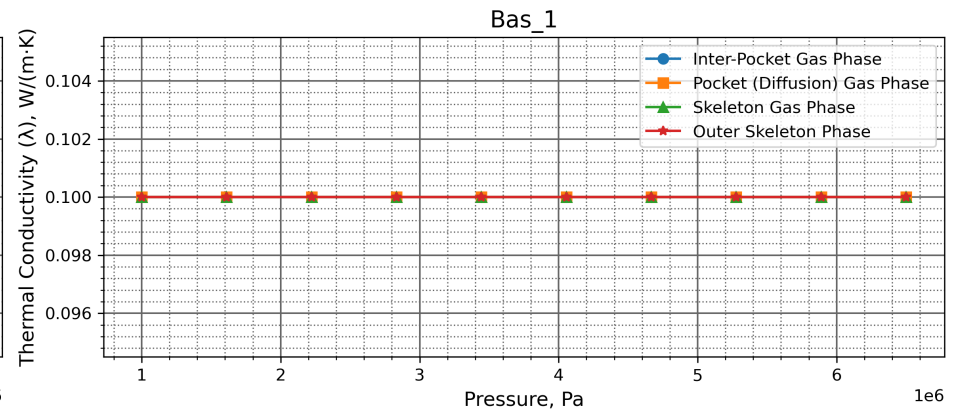
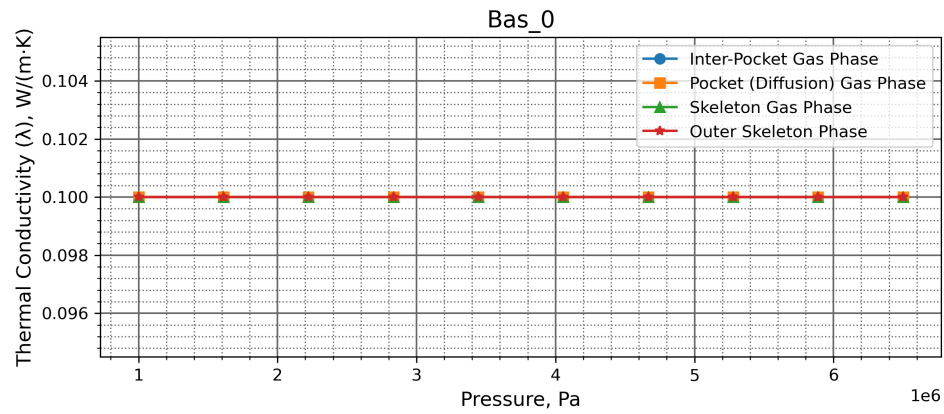
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

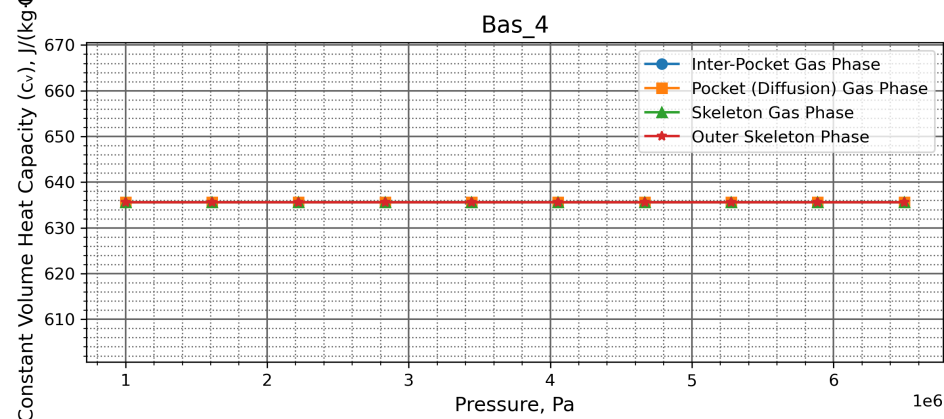
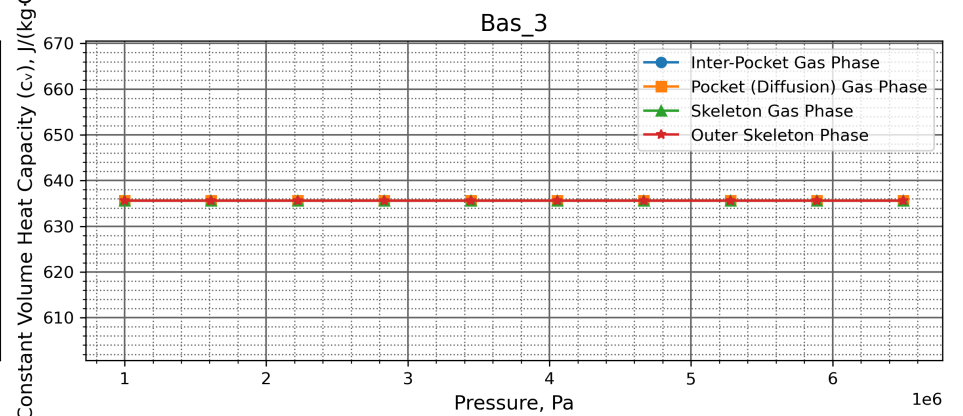
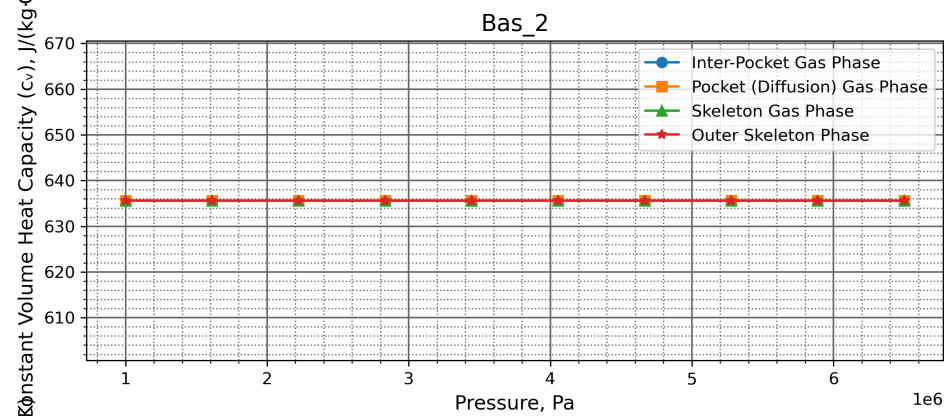
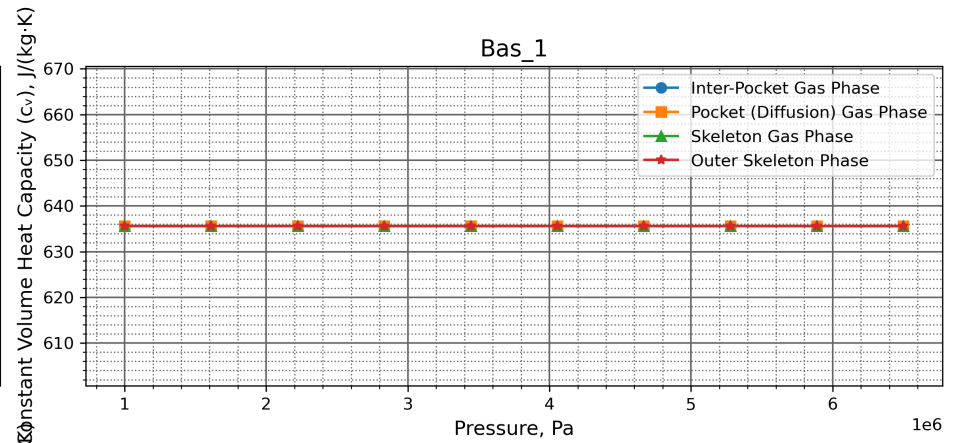
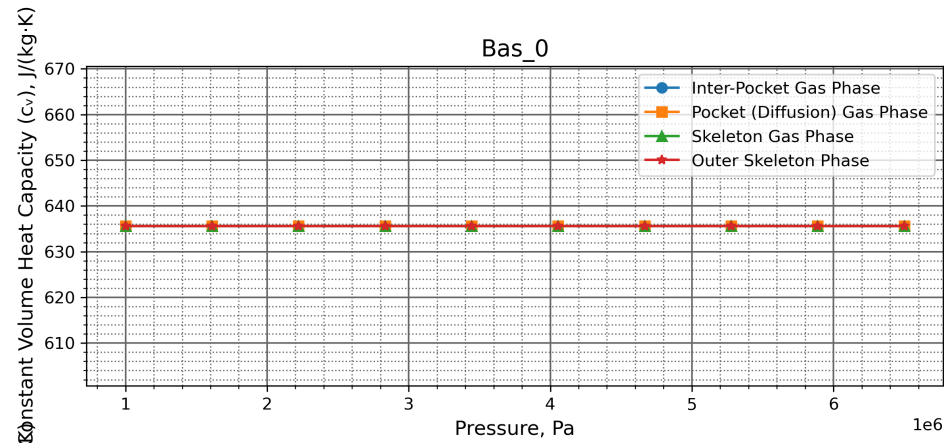
Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

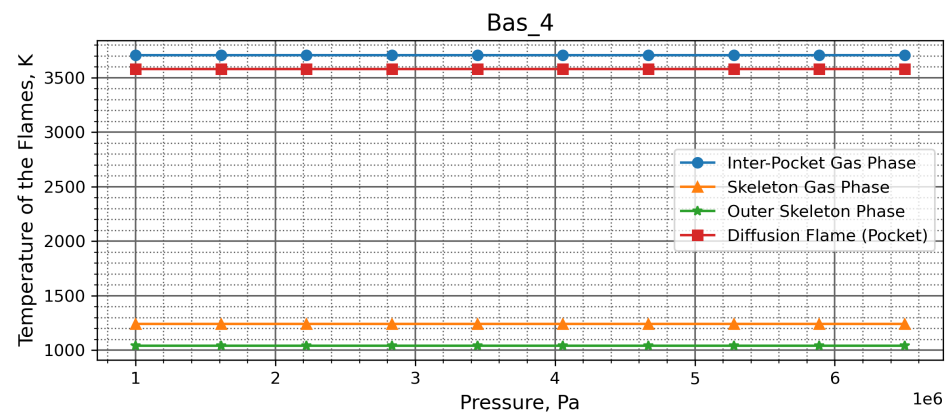
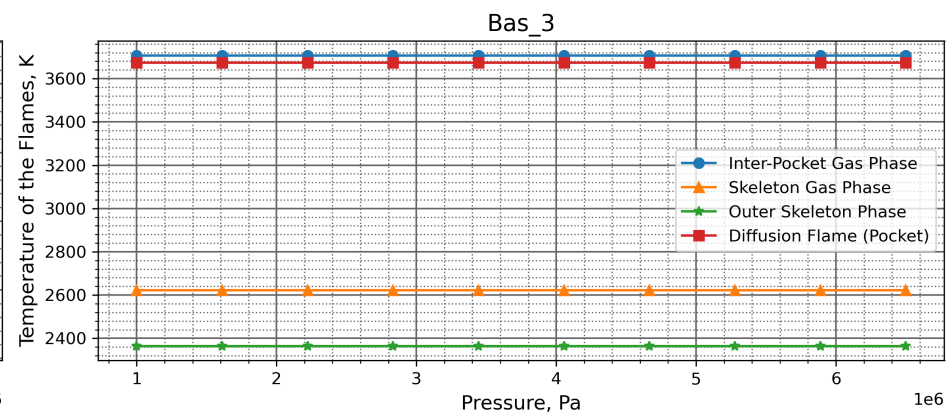
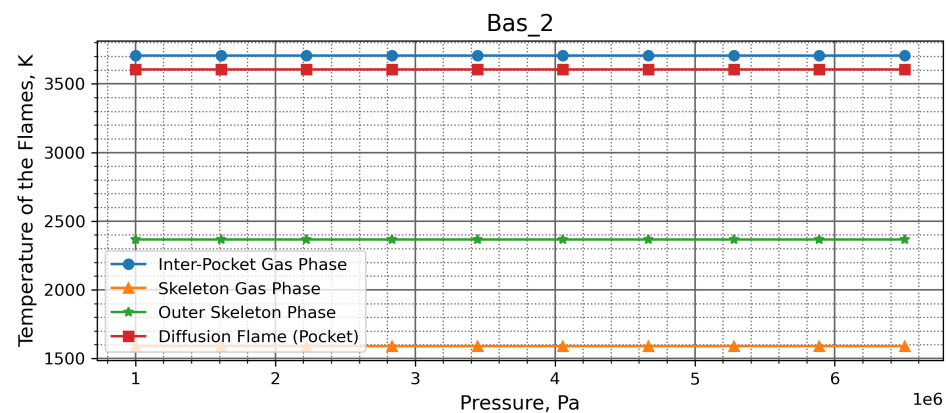
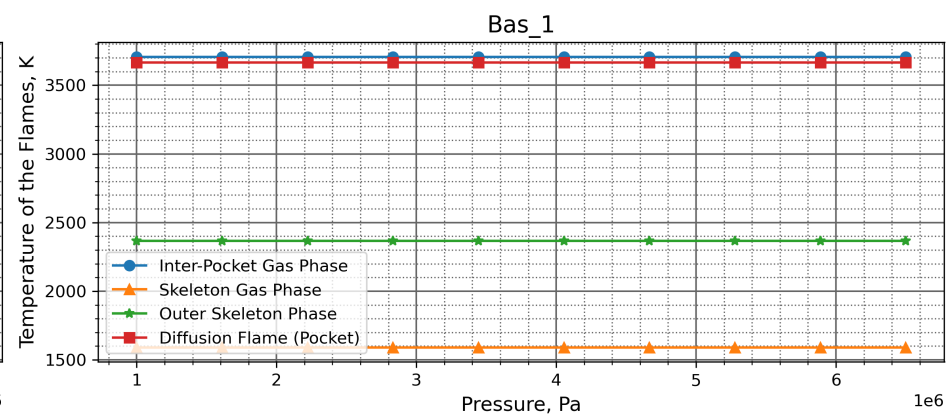
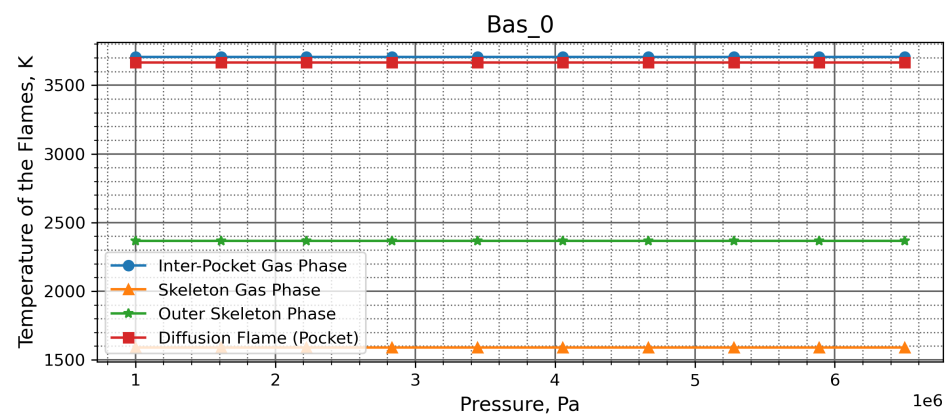
Thermal Conductivity (λ), W/(m·K)



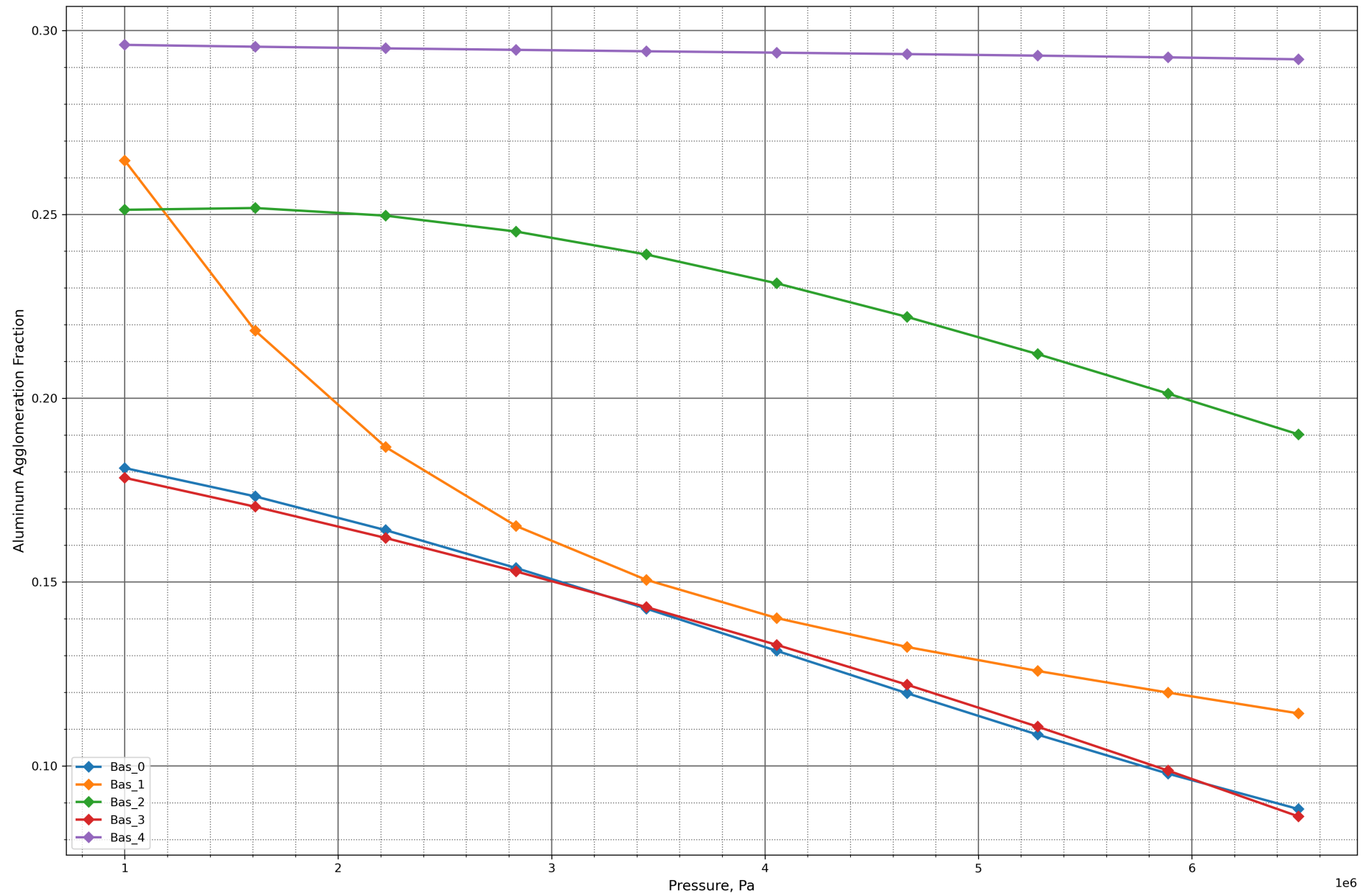
Constant Volume Heat Capacity (c_v), J/(kg·K)



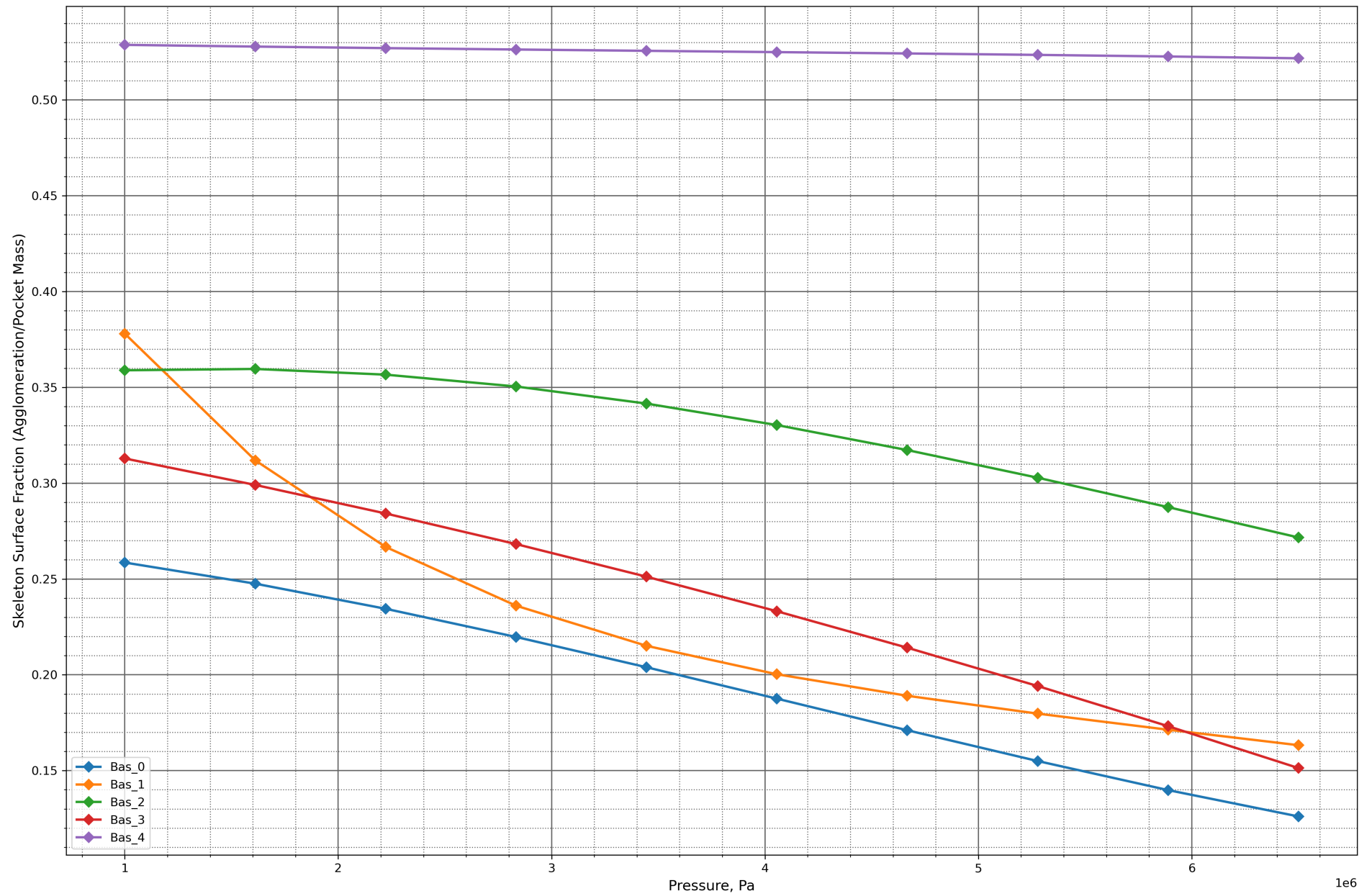
Temperature of the Flames, K



Aluminum Agglomeration Fraction



Skeleton Surface Fraction (Agglomeration/Pocket Mass)



Performance Report

System Information

Operating System: Unix 6.8.0.124 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.9 (.NET 10.0.9)
Architecture: X64 / X64

Total Execution Time

Total execution time: 02:45:29.416

Garbage Collection Information

Total Memory: 8556448 bytes
GC Pause Time Percentage: 0.14%
Generation 0 Collections: 12186
Generation 1 Collections: 982
Generation 2 Collections: 99

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 6814373

Maximum execution time: 23.6343 ms

Minimum execution time: 0.0001 ms

Mean execution time: 0.7865717937951674 ms

Execution time standard deviation: 0.24675720214692984 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 5780154

Maximum execution time: 19.3394 ms

Minimum execution time: 0.0001 ms

Mean execution time: 0.24657745092946362 ms

Execution time standard deviation: 0.11062779690218126 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 4843485

Maximum execution time: 23.39 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2803052409163953 ms

Execution time standard deviation: 0.07002579104947027 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 20.412 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7902476047639938 ms

Execution time standard deviation: 0.24850385660541782 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 5659398

Maximum execution time: 15.411 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.24774306747817845 ms

Execution time standard deviation: 0.1125080487022167 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 4706608

Maximum execution time: 20.6015 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.2821972848174256 ms
Execution time standard deviation: 0.06873460858329537 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 6713952
Maximum execution time: 34.1129 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7876286149051491 ms
Execution time standard deviation: 0.2521314808408164 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 5635750
Maximum execution time: 20.3701 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2473149380118196 ms
Execution time standard deviation: 0.11237826385703215 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 4683292
Maximum execution time: 14.1347 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28212978208488276 ms
Execution time standard deviation: 0.06978191203792686 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 6713952
Maximum execution time: 28.59 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7929331111244394 ms
Execution time standard deviation: 0.24948744924995692 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 5651484
Maximum execution time: 17.9092 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2472953480360421 ms
Execution time standard deviation: 0.11201091833499088 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 4689203
Maximum execution time: 21.1051 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28290493237340614 ms
Execution time standard deviation: 0.0677486659592496 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 6713952
Maximum execution time: 32.4252 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.784312500402153 ms
Execution time standard deviation: 0.2521098942132896 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 5623354
Maximum execution time: 30.3454 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24611085928434112 ms
Execution time standard deviation: 0.11351230059187468 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 4662229

Maximum execution time: 16.0563 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28141053826399 ms

Execution time standard deviation: 0.0688886525939947 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 35.9331 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.786568784808213 ms

Execution time standard deviation: 0.2505493852682122 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 5661179

Maximum execution time: 14.3247 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2460379785553794 ms

Execution time standard deviation: 0.1119157382547391 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 4706828

Maximum execution time: 21.5605 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.281304083769355 ms

Execution time standard deviation: 0.0695946976878418 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 33.6796 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7919298615479171 ms

Execution time standard deviation: 0.2465876737784819 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 5651072

Maximum execution time: 21.2221 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.24877925252058317 ms

Execution time standard deviation: 0.11101202755062935 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 4706954

Maximum execution time: 19.9639 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2835191016313103 ms

Execution time standard deviation: 0.06783414304936464 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 25.6073 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7857128824126263 ms

Execution time standard deviation: 0.2507951140224262 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 5628502

Maximum execution time: 24.9362 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24549259119035285 ms
Execution time standard deviation: 0.11309164872642707 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 4657719

Maximum execution time: 15.1964 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2817670239230714 ms
Execution time standard deviation: 0.06775368735297359 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 34.3939 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7881574324481125 ms
Execution time standard deviation: 0.250164945027672 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 5631171

Maximum execution time: 13.5471 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24782636400136138 ms
Execution time standard deviation: 0.11151799568717388 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 4678132

Maximum execution time: 20.7571 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2828407745869573 ms
Execution time standard deviation: 0.06878821389946792 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 18.0048 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7885806592152941 ms
Execution time standard deviation: 0.24747218310299715 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 5667600

Maximum execution time: 25.3094 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.247500753264199 ms
Execution time standard deviation: 0.11246829938485674 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 4721196

Maximum execution time: 18.3366 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28195930736618613 ms
Execution time standard deviation: 0.06794847432235307 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 18.3012 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7891285457060963 ms

Execution time standard deviation: 0.24864144078022676 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 5641097

Maximum execution time: 22.3335 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2480405240682665 ms

Execution time standard deviation: 0.11242228503013034 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 4691640

Maximum execution time: 34.7682 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2831129570469985 ms

Execution time standard deviation: 0.07061159637889232 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 31.943 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7919086084917009 ms

Execution time standard deviation: 0.24936868594288464 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 5632131

Maximum execution time: 19.8012 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2478310857116139 ms

Execution time standard deviation: 0.11302182416610346 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 4668472

Maximum execution time: 22.1003 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28432667735825934 ms

Execution time standard deviation: 0.06755002270249205 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 37.7023 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7901881050089025 ms

Execution time standard deviation: 0.2510632347994322 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 5642044

Maximum execution time: 18.2007 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.24737232012014915 ms

Execution time standard deviation: 0.1123392606396133 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 4687192

Maximum execution time: 16.6351 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28220043834349046 ms

Execution time standard deviation: 0.06686924499211729 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 6713952
Maximum execution time: 32.7912 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7894168682916423 ms
Execution time standard deviation: 0.24991906750090156 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 5645810
Maximum execution time: 25.0589 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24742464252249888 ms
Execution time standard deviation: 0.11249448319996135 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 4694076
Maximum execution time: 25.2625 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28252133567928733 ms
Execution time standard deviation: 0.0689159243214756 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 6713952
Maximum execution time: 26.2661 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7852971231548751 ms
Execution time standard deviation: 0.2515717724685557 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 5646143
Maximum execution time: 22.4432 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2449768460876694 ms
Execution time standard deviation: 0.11336636597247385 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 4670044
Maximum execution time: 21.6655 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2802820001267307 ms
Execution time standard deviation: 0.0704091827396997 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 6713952
Maximum execution time: 25.2453 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7937272528162704 ms
Execution time standard deviation: 0.24652767372598097 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 5648441
Maximum execution time: 33.1793 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24826710826937354 ms
Execution time standard deviation: 0.11303205947711616 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 4684510
Maximum execution time: 15.4785 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.2840704761010431 ms
Execution time standard deviation: 0.06710410532308014 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance

Call count: 6713952
Maximum execution time: 26.5889 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7900532881676986 ms
Execution time standard deviation: 0.25091684593853386 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance

Call count: 5631608
Maximum execution time: 15.5019 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24720746182972858 ms
Execution time standard deviation: 0.11252971995224581 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance

Call count: 4667342
Maximum execution time: 18.963 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28314420768823484 ms
Execution time standard deviation: 0.06880466837794583 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance

Call count: 6713952
Maximum execution time: 46.0902 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7888265203415694 ms
Execution time standard deviation: 0.25027800100892905 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance

Call count: 5639470
Maximum execution time: 15.0443 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24851369992218436 ms
Execution time standard deviation: 0.11148970469634288 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance

Call count: 4703656
Maximum execution time: 12.844 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28349058183674825 ms
Execution time standard deviation: 0.06698302061293918 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance

Call count: 6713952
Maximum execution time: 19.0357 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.795818788904091 ms
Execution time standard deviation: 0.2464010335374565 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance

Call count: 5658089
Maximum execution time: 16.615 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2492118552924826 ms
Execution time standard deviation: 0.11235527459310574 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 4701875
Maximum execution time: 18.1041 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28494469989097776 ms
Execution time standard deviation: 0.06692545457165062 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 6713952
Maximum execution time: 26.922 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.79478211929424 ms
Execution time standard deviation: 0.24541446681664408 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 5667395
Maximum execution time: 27.3629 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.25118934852781927 ms
Execution time standard deviation: 0.11082576862529864 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 4751717
Maximum execution time: 17.6302 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28479199670354516 ms
Execution time standard deviation: 0.06617700402104545 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 6713952
Maximum execution time: 29.8865 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7930694316402739 ms
Execution time standard deviation: 0.24629470962971353 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 5663483
Maximum execution time: 18.5178 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24966883931673697 ms
Execution time standard deviation: 0.11050081586837744 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 4731207
Maximum execution time: 23.8978 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2833864420009987 ms
Execution time standard deviation: 0.06758670876343965 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 6713952
Maximum execution time: 27.6228 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7880626418538965 ms
Execution time standard deviation: 0.24976056927310755 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 5652841
Maximum execution time: 12.3043 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.24689384070773068 ms
Execution time standard deviation: 0.1116922752075937 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 4693544
Maximum execution time: 20.9466 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2815952691612118 ms
Execution time standard deviation: 0.07124828616384589 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 6713952
Maximum execution time: 28.0087 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7885702728586829 ms
Execution time standard deviation: 0.24907658368693167 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 5643543
Maximum execution time: 20.5422 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24692642354279298 ms
Execution time standard deviation: 0.11192028126632292 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 4686880
Maximum execution time: 15.3431 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2822960455356256 ms
Execution time standard deviation: 0.06650289873664468 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 6713952
Maximum execution time: 19.5664 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7926911551943558 ms
Execution time standard deviation: 0.247276796853111 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 5659906
Maximum execution time: 18.4681 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24841606295229832 ms
Execution time standard deviation: 0.11118253802074794 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 4711745
Maximum execution time: 14.886 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2827319648664915 ms
Execution time standard deviation: 0.0666580183053647 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 6713952
Maximum execution time: 32.533 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7828005755328795 ms
Execution time standard deviation: 0.25090623753899044 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 5641780

Maximum execution time: 13.9455 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.24491790489880066 ms

Execution time standard deviation: 0.11199935293439604 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 4679415

Maximum execution time: 53.3131 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28046982610005144 ms

Execution time standard deviation: 0.07202015353599714 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 17.7778 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7870830824080328 ms

Execution time standard deviation: 0.24785176897744188 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 5662605

Maximum execution time: 18.0756 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.24670339146031092 ms

Execution time standard deviation: 0.11051822187730817 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 4717224

Maximum execution time: 11.9546 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28047604444898433 ms

Execution time standard deviation: 0.06664669244807633 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 25.0173 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7960535361438601 ms

Execution time standard deviation: 0.24496827681610636 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 5659759

Maximum execution time: 13.3511 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2496572815379396 ms

Execution time standard deviation: 0.11022872012329568 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 4721896

Maximum execution time: 36.3474 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28441294920515714 ms

Execution time standard deviation: 0.06729040190060663 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 22.9595 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7903562476615311 ms
Execution time standard deviation: 0.24847548969184025 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 5654610
Maximum execution time: 12.1292 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24758056209005014 ms
Execution time standard deviation: 0.11074597962898107 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 4713061
Maximum execution time: 9.6185 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28209752118631826 ms
Execution time standard deviation: 0.06631514783412601 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 6713952
Maximum execution time: 18.6696 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7873821683712323 ms
Execution time standard deviation: 0.24797523431327753 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 5639296
Maximum execution time: 26.2022 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24682512680663862 ms
Execution time standard deviation: 0.11140355536597335 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 4680516
Maximum execution time: 24.3588 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2830066374519219 ms
Execution time standard deviation: 0.06885663303454445 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 6713952
Maximum execution time: 27.9097 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.794378970329263 ms
Execution time standard deviation: 0.24447383673937403 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 5683259
Maximum execution time: 10.3445 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.24845031963173458 ms
Execution time standard deviation: 0.10997145475404861 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 4737524
Maximum execution time: 25.9155 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28335654903276825 ms

Execution time standard deviation: 0.06852393941470562 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 25.9439 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7866789194352379 ms

Execution time standard deviation: 0.24829612627994388 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 5659501

Maximum execution time: 20.3889 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2475981786203387 ms

Execution time standard deviation: 0.11213705695998677 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 4708326

Maximum execution time: 15.7611 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2816639706978761 ms

Execution time standard deviation: 0.0670975442179559 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 22.3182 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7871736168206007 ms

Execution time standard deviation: 0.246911273693169 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 5651200

Maximum execution time: 24.3244 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.24766493024491085 ms

Execution time standard deviation: 0.1100185278116964 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 4707995

Maximum execution time: 32.1622 ms

Minimum execution time: 0.0003 ms

Mean execution time: 0.28181849456083874 ms

Execution time standard deviation: 0.06693696967129531 ms

Method: Telemetry/Proc/32/OptimizationProblemExecutionPerformance

Call count: 6713952

Maximum execution time: 32.7042 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7884813021451085 ms

Execution time standard deviation: 0.2489055157192164 ms

Method: Telemetry/Proc/32/OptimizationProblemExecutionPerformance

Call count: 5671347

Maximum execution time: 21.1046 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.24655854824258827 ms

Execution time standard deviation: 0.112244801228024 ms

Method: Telemetry/Proc/32/OptimizationProblemExecutionPerformance

Call count: 4721907
Maximum execution time: 25.3842 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2813159464809552 ms
Execution time standard deviation: 0.06765742184764255 ms